

Warehouse Management System

Man-Days Estimation — Chiffrage

Optimistic scenario — senior developer (× 5/8)

Code written by Claude Opus 4.8, implemented by an experienced senior developer

Scope: New WMS module for a 3PL (ERP back-office + PWA scanner), integrated into the existing host

Basis: Bottom-up over 60 build items across 9 phases, calibrated to the agreed mockups

Scenario: Optimistic — the baseline Opus 4.8 estimate × 5/8 (0.625) for a senior developer

Status: Optimistic planning estimate, for client review

Date: 19 June 2026

1 · Headline Estimate

Estimated effort — Claude Opus 4.8 with a senior developer (optimistic): **≈ 229 man-days** (planning / P50).

With a 25% contingency for the brownfield-integration unknowns: **≈ 287 man-days** (P80 — the figure to budget to).

This optimistic scenario applies a **5/8 (0.625) senior-developer productivity factor** to the baseline Opus 4.8 estimate of ≈ 364 / ≈ 456 man-days (P50 / P80), reflecting an experienced senior developer implementing the application.

For comparison, the same scope hand-coded by a traditional team is ≈ 970–1212 man-days. **Opus 4.8 with a senior developer removes roughly 76% of that effort** — the code is generated; the remaining man-days are human specification, review, host integration, testing and iteration, at a senior pace.

All figures are human man-days (≈ one person-day each), not calendar time. Opus 4.8 runtime is a tool cost, not a man-day. See §2 for the method and assumptions.

Summary by phase

#	Phase	Items	Traditional md	Opus 4.8 + Sr md
0	Foundation / Kernel	10	122	29
1	Master Data	8	59.5	12
2	Goods Reception	7	95	23.3
3	Putaway	3	43	10.7
4	Stock Visibility & Status	3	30	6.9
5	Stock-Out (Outbound)	8	116	28.9
6	Inventory Operations	8	108	26.4
7	Reports & Visibility	7	68.5	15.2
8	Hardening & Closure	6	66	15.1
	Engineering subtotal	60	708	167.5

Roll-up to the budget figure

Roll-up	Traditional md	Opus 4.8 + Sr md
Engineering (sum of all build items)	708	167.5
+ Delivery overhead (PM, system test & UAT, iteration, release) — 37%	262	62
= Project subtotal (P50, planning)	970	229
+ Risk contingency (brownfield unknowns) — 25%	242	57
= Total to budget (P80)	1212	287
Optional — PWA offline sync (only if a v1 requirement, decision #5)	+ 48	+ 13

The optional offline-sync line is excluded from the totals above; add it only if offline scanning is confirmed as a v1 requirement (it is the single largest cost swing).

2 · Method & Assumptions

What a man-day means here

A man-day is one person-day (~8 h) of human engineering effort. Because Claude Opus 4.8 writes the code, that day is spent specifying the change, directing the model, reviewing and correcting what it generates, wiring it into the existing host, testing, and iterating with the client — not hand-typing the bulk of the code. This optimistic scenario further assumes an experienced senior developer fluent in the host stack (.NET Core 3.1 / Angular 14 / SQL Server 2014), who reviews, integrates and decides faster than an average engineer.

How each figure is built

Every build item is sized on the project estimation guide's complexity tiers (which sum backend, database, ERP UI, PWA UI and test effort) to give a traditional human figure. An Opus-4.8 acceleration factor is applied per tier to get the baseline Opus figure; this optimistic scenario then multiplies that by a 5/8 (0.625) senior-developer factor:

Tier	Profile	Trad. md	Opus 4.8 md	× 5/8	Final md
S	CRUD master over the generic base service	4	1.0	0.625	0.6
S–M	small CRUD with a little logic	6.5	2.0	0.625	1.3
M	CRUD + non-trivial rules / one guard	9	3.0	0.625	1.9
M–L	multi-step, some cross-cutting calls	13	5.0	0.625	3.1
L	multi-step dual-channel flow, 3+ cross-cutting calls	17	7.0	0.625	4.4
XL	engine / orchestration, heavy edge cases	28	12.5	0.625	7.8

“Opus 4.8 md” are the figures from the standard estimate; “Final md” = those × 5/8, rounded per item. Tier baselines are calibrated against the project’s estimation guide and the completed mockups as a reference class.

What is included

- Backend domain + services + APIs, database schema & migrations (SQL Server 2014), ERP screens (Angular/NgRx/Kendo), PWA scanner screens, and per-item tests.
- Integration into the existing host — Swagger client regeneration, gateway routing, DI wiring, identity/role reuse.
- Delivery overhead — project management, system & integration testing, client UAT support, environment/CI-CD, SQL migration runbook and release.
- A risk contingency keyed to the brownfield unknowns (SQL 2014 child-table modelling, the shared Auth module, .NET 3.1 / Angular 14 tooling, the open decisions).

What is excluded (price separately)

- Client portal and billing (out of v1 by decision).
- Label-printer / hardware integration and procurement (the mock prints client-side; production needs a print service).
- Infrastructure & hosting, data migration from any legacy WMS, end-user training & change management, and post-go-live support / SLA.

Key assumptions

- Optimistic scenario: an experienced senior developer (the 5/8 factor). Treat the standard Opus 4.8 estimate (≈ 364 / ≈ 456 md) as the more conservative companion figure.
- The functional scope and UX are locked — the Phase-2 mockups and the docs/ specs act as the specification, which removes most analysis/design effort (already reflected in the factors).
- Brownfield build: a new WMS module inside the existing ERP, reusing identity, data-access, gateway and UI kit.
- Online-only PWA for v1. Offline scanning (decision #5) is the separate optional line.
- Stay on the host runtimes (.NET Core 3.1 / Angular 14 / SQL Server 2014); EF migrations for schema; WMS-side tenant scoping.
- One delivery team. Figures are human man-days, not elapsed calendar time (parallelism shortens the calendar, not the man-days).

3 · Detail by Phase & Functionality

One line per build item. "Delivers" names the function or page(s) produced. Traditional = hand-coded human days; Opus 4.8 + Sr = senior-developer days with the model writing the code (the baseline Opus figure × 5/8).

Phase 0 · Foundation / Kernel

One-time enabling platform — module, schema, and the shared services every process reuses.

Item	Delivers	Tier	Trad. md	Opus+Sr md
P00-S01	Backend module skeleton (7 projects) + DI / gateway route / Swagger client pipeline; Angular ERP + PWA project shells & routing	M	9	1.9
P00-S02	SQL Server 2014 schema foundation, EF DbContext, audit-ledger table, migrations + seed from the mock dataset	L	17	4.4
P00-S03	Tenant / site scoping service + WMS role/claim integration (touches the shared Auth module)	L	17	4.4
P00-S04	Stock-ledger service — append-only transaction log, written in-transaction by every action	M	9	1.9
P00-S05	Freeze service — the cross-screen frozen-scope guard	M	9	1.9
P00-S06	Capacity + segregation service — single source of truth for every bin write	M	9	1.9
P00-S07	Allocation engine core — FEFO / FIFO + reservation-aware availability	L	17	4.4
P00-S08	Reason-code service + approval / maker-checker framework	M	9	1.9
P00-S09	Work-item assignment facet (My-tasks / claim) + tracking-flag (lot/expiry/serial) plumbing	M	9	1.9
P00-S10	PWA shell — scan engine (camera / ZXing / manual / list), service worker, shared models & API client	L	17	4.4
Phase 0 subtotal			122	29

Phase 1 · Master Data

Mostly the CRUD recipe over the Phase-0 base classes — highly parallel.

Item	Delivers	Tier	Trad. md	Opus+Sr md
P01-S01	Clients (+ back-order policy), Suppliers, Carriers, Consignees	S-M	6.5	1.3
P01-S02	Sites (+ addressing levels, storage areas), Locations (+ capacity, area)	M	9	1.9

Item	Delivers	Tier	Trad. md	Opus+Sr md
P01-S03	Products (+ tracking flags, category, weight, preferred storage), Categories, UoM	M	9	1.9
P01-S04	Packaging — shared templates + per-product packaging with per-level barcodes	M	9	1.9
P01-S05	Reason Codes master (all domains)	S	4	0.6
P01-S06	Users + WMS roles/rights + client/site scope + client–user mapping	M	9	1.9
P01-S07	PWA lookups (location, product) — read-only	S	4	0.6
P01-S08	Bulk CSV import (products, locations)	M	9	1.9
	Phase 1 subtotal		59.5	12

Phase 2 · Goods Reception

First stock-creating process — ASN → receive → inspect → GRN, minting LPNs. Dual-channel.

Item	Delivers	Tier	Trad. md	Opus+Sr md
P02-S01	ASN management — CRUD, lines, derived open / partial / closed status	M	9	1.9
P02-S02	Receive against ASN + blind → mint LPNs, LPN split (per-each / per-pack), close-short, wrong-client guard	L	17	4.4
P02-S03	Mixed-pallet build (aggregate LPN + manifest label) + decomposition hand-off to Putaway	L	17	4.4
P02-S04	Inspection — quantity-based partial accept/reject, dispositions, reason codes	L	17	4.4
P02-S05	GRN — Goods Received Note (document / print, one per receipt)	M	9	1.9
P02-S06	Labels / reprint (LPN + pallet, recovery by ASN / recent, audited)	M	9	1.9
P02-S07	Reception edge cases — refuse-at-door, over-receipt approval, duplicate / damaged / expired guards, cold-chain	L	17	4.4
	Phase 2 subtotal		95	23.3

Phase 3 · Putaway

Directed slotting into bins, enforcing capacity / segregation / freeze. Dual-channel.

Item	Delivers	Tier	Trad. md	Opus+Sr md
P03-S01	Single-LPN directed putaway — ranked bins (home→consolidate→category→open), guards, partial split	L	17	4.4
P03-S02	Mixed-pallet decomposition — operator-chosen order → child LPNs	L	17	4.4
P03-S03	Damage-found reject, overflow-park, resume / partial-progress visibility	M	9	1.9
	Phase 3 subtotal		43	10.7

Phase 4 · Stock Visibility & Status

The read model + status engine that Stock-Out and Reports both depend on.

Item	Delivers	Tier	Trad. md	Opus+Sr md
P04-S01	Stock-on-hand / availability read model + query API (the hot path for outbound & reports)	L	17	4.4
P04-S02	Stock Status screen + reason-coded status-change engine	M	9	1.9
P04-S03	Expose blocked / frozen / expiry / in-transit flags consistently across every stock payload	S	4	0.6
	Phase 4 subtotal		30	6.9

Phase 5 · Stock-Out (Outbound)

The most complex process — orders → allocation → pick/dispatch → delivery notes. Dual-channel.

Item	Delivers	Tier	Trad. md	Opus+Sr md
P05-S01	Outbound orders — CRUD, lines, ship-to consignee, full-stock-out, cancel order / line, status flow	M-L	13	3.1
P05-S02	Allocation surface — FEFO/FIFO, reservation-aware, short-pick, manual override	L	17	4.4
P05-S03	Pick / Dispatch (classic) — scan-confirm, serials-on-issue, carrier + proof-of-load, Delivery Note	L	17	4.4
P05-S04	Express Fulfil — one-pass allocate→pick→dispatch, role-gated	L	17	4.4
P05-S05	Back-order vs short-close + multi-shipment delivery notes	M	9	1.9
P05-S06	Ad-hoc / emergency dispatch — governance gate, role-gated + back-office approval queue	L	17	4.4
P05-S07	Return-to-vendor / client — non-consignee outbound	M	9	1.9

Item	Delivers	Tier	Trad. md	Opus+Sr md
P05-S08	PWA pick / dispatch + ad-hoc + RTV parity	L	17	4.4
	Phase 5 subtotal		116	28.9

Phase 6 · Inventory Operations

In-warehouse stock operations incl. the freeze workflow; each card largely standalone.

Item	Delivers	Tier	Trad. md	Opus+Sr md
P06-S01	Move — intra-site relocation, partial split, guards	M–L	13	3.1
P06-S02	Transfer — inter-site ship → in-transit → receive	L	17	4.4
P06-S03	Cycle Count — multi-location sheet → approve corrects stock; found / missing	L	17	4.4
P06-S04	Physical Inventory — freeze → count → post → unfreeze	L	17	4.4
P06-S05	Adjustments (qty) + Corrections (attributes) — approval-gated	M	9	1.9
P06-S06	Repack / split / merge / re-kit — genealogy + balance guard	L	17	4.4
P06-S07	Returns / put-back — dispositions (restock / via-putaway / quarantine)	M	9	1.9
P06-S08	Disposal / Scrap — approval-gated, terminal, maker-checker	M	9	1.9
	Phase 6 subtotal		108	26.4

Phase 7 · Reports & Visibility

Read-only over the stock read model + the ledger. ERP only.

Item	Delivers	Tier	Trad. md	Opus+Sr md
P07-S01	Stock on Hand	M	9	1.9
P07-S02	Transaction History / stock ledger	M	9	1.9
P07-S03	Expiry & Aging	S–M	6.5	1.3
P07-S04	Inbound & Outbound activity	M	9	1.9
P07-S05	Variance & Bin / Location Utilization	M	9	1.9

Item	Delivers	Tier	Trad. md	Opus+Sr md
P07-S06	Lot / Serial Traceability + Stock Card / Product Ledger	L	17	4.4
P07-S07	Client Stock Statement	M	9	1.9
	Phase 7 subtotal		68.5	15.2

Phase 8 · Hardening & Closure

Production-grade non-functionals + carry-overs. (PWA offline sync is a separate optional line.)

Item	Delivers	Tier	Trad. md	Opus+Sr md
P08-S01	Export (CSV / Excel) across all reports	M	9	1.9
P08-S02	Performance pass — indexes, query tuning, load tests on SQL 2014, grid paging	M–L	13	3.1
P08-S03	Security review — server-side scoping on every endpoint, role matrix, authz tests	L	17	4.4
P08-S04	Audit completeness — immutable ledger + before/after snapshots, reprint / override auditing	M	9	1.9
P08-S05	Operations dashboard wired to live rollups	M	9	1.9
P08-S07	Carry-over alignment sweep — shared capacity service, dispatch-txn parity, etc.	M	9	1.9
	Phase 8 subtotal		66	15.1

Optional — conditional on decision #5

Item	Delivers	Tier	Trad. md	Opus+Sr md
P08-S06	PWA offline sync — IndexedDB queue, sync-on-reconnect, conflict resolution (only if offline scanning is a v1 requirement — open decision #5)	XL	28	7.8

4 · Overhead, Risk & Adjusting the Estimate

Delivery overhead

Applied as a percentage of the engineering subtotal (it scales with the work):

Overhead component	%	Trad. md	Opus+Sr md
System test, integration & UAT	12%	85	20.1
Project management & coordination	12%	85	20.1
Client review & iteration loops	8%	56.6	13.4
Environment, CI/CD, SQL migration & release	5%	35.4	8.4
Total overhead	37%	262	62

Risk contingency

A flat 25% contingency takes the planning (P50) figure to a budget (P80) figure. The risk is concentrated in the brownfield integration — modelling the mock's nested data as SQL Server 2014 child tables, extending the shared Auth module for tenant/site scoping, the EOL .NET Core 3.1 / Angular 14 tooling, and resolving the five open integration decisions. Note that an optimistic productivity factor and a thin contingency compound: if the senior-developer pace does not hold, this scenario has less cushion than the standard estimate.

How to adjust

- The estimate is a model with five levers: the per-tier baseline days, the Opus-4.8 factor per tier, the senior-developer factor (here 5/8), the overhead %, and the risk %. Change any one and the totals re-flow.
- The senior-developer factor (5/8) is the optimistic lever in this version. Setting it to 1.0 returns the standard Opus 4.8 estimate ($\approx 364 / \approx 456$ md); a value between the two brackets the realistic range.
- Recalibrate the factors after the first 2–3 real build items — the actual senior pace on this host will quickly show whether 5/8 holds, and the rest of the estimate follows.
- A line-by-line, adjustable spreadsheet version (one row per build item, with all the levers on an assumptions sheet) is the natural next artifact if you want to tune the numbers interactively.

This is an optimistic, first-cut planning estimate. It is not a fixed-price quotation; budget against the P80 total in §1, and keep the standard Opus 4.8 estimate as the more conservative companion.